

Malama Masheke

+1(704)223-7645 | gabrielmasheke@gmail.com

malamamasheke.com | github.com/malgmash | linkedin.com/in/malama-masheke

EDUCATION

Livingstone College

B.S. Computer Information Systems

Expected Graduation: December 2029

GPA: 4.0/4.0

Relevant Coursework: Computer Systems, Intro to Programming, Calculus, Intro to Computer Science (Harvard CS50), Python: Machine Learning and Deep Learning (Udemy)

TECHNICAL SKILLS

- **Languages:** Python, C, C++, Java, JavaScript/TypeScript, SQL, MATLAB
- **Tools:** Codex, Claude Code, Git, Linux, GitHub, Bash, VS Code, Jupyter Notebook, Google Colab, Cursor
- **Foundations:** API Design, Client-Server Architecture, Query Design, Machine Learning and Deep Learning

TECHNICAL LEADERSHIP

Dental Time Machine | AI Dental Cost Planner

October 2026

<https://dental-time-machine-965s.vercel.app/>

- **Technical Infrastructure:** Python, FastAPI, Amazon Bedrock, Claude, Amazon Polly, ElevenLabs, React, TypeScript, Tailwind CSS, pytest, Vercel, System Design, Optimization, Automated Testing
- Led a five-person team in a 20-hour hackathon to ship a web app that turns confusing dental treatment plans into plain answers on what insurance pays, what the patient pays, and how rescheduling care can lower the final bill.
- Architected the Python FastAPI backend around one rule, the AI explains but never calculates, so a cost engine verified by 1,000+ automated tests produces every dollar figure and an optimizer finds the cheapest allowed schedule.
- Integrated an AI agent using Claude on Amazon Bedrock that reads uploaded documents and talks with users in four languages, plus a guard blocking any AI-stated dollar amount the engine did not produce, keeping every number accurate.

AuditX | Financial Auditing Platform for Companies

September 2026

<https://audit-x-three.vercel.app>

- **Technical Infrastructure:** Next.js 15, TypeScript, PostgreSQL, Prisma, pgvector, Python, FastAPI, NVIDIA NIM, Auth.js, Server-Sent Events, Recharts, Vercel
- Led product and backend for a full-stack web app built in a 24-hour hackathon that audits employee expenses, receipts, and timesheets at 20 to 100 person companies, flagging possible misuse of company funds for admin review.
- Designed two services sharing one PostgreSQL database, a Next.js app for the interface and a Python FastAPI service that runs three fraud checks spotting duplicate receipts, unusual spending, and impossible timesheet hours.
- Applied statistics against each company's own spending history to score risk, used an NVIDIA NIM AI model to write plain-English case summaries, and removed personal data before any AI call, leaving final decisions to a human admin.

TECHNICAL PROJECTS

Dēmos | Voting Platform with Data-Layer Authorization

July 2026

demos-rust-eight.vercel.app

- **Technical Infrastructure:** TypeScript, Next.js, Node.js, PostgreSQL via Supabase, Row-Level Security, SQL migrations, Supabase Auth and Realtime, Vercel
- Deployed an 11,700-line full-stack voting platform where organizations run private elections in single, multiple, or ranked-choice formats, designing the database schema and system design document before writing any code.
- Engineered security into the database layer with 40 Row-Level Security policies across 12 tables, rules that let each user reach only data they are allowed to see, and authored 17 SQL migrations to evolve the schema safely.
- Programmed the Instant-Runoff Voting algorithm that counts ranked ballots round by round, closed a ballot-secrecy gap that could link votes to voters, and directed Claude Code as an AI coding partner while reviewing each change.

Tri-Ask-Me | AI Support Triage Agent

May 2026

<https://github.com/malgmash/Tri-Ask-Me>

- **Technical Infrastructure:** Amazon Bedrock, Retrieval-Augmented Generation (RAG), TF-IDF search, prompt engineering, modular agent pipeline
- Shipped a terminal-based AI agent in 24 hours as a hackathon project that triages real customer support tickets across HackerRank, Claude, and Visa, replying to routine questions and routing sensitive ones to human specialists.
- Implemented a four-stage pipeline (retrieve, classify, respond, orchestrate) that uses RAG, searching a library of help articles first, so the Amazon Bedrock AI model answers from real company documentation instead of guessing.
- Added safeguards that refuse to answer when no source is found and route 30+ keyword triggers covering security, financial, legal, and privacy cases to human review, a safety-first design that earned a top 10% global ranking.

LEADERSHIP AND COMMUNITY IMPACT

- **Adobe Student Ambassador | Present:** Selected to represent Adobe at Livingstone College, running live demos and workshops that help classmates adopt Adobe Express and Creative Cloud, and partnering with student groups to reach more users.